

Programs on Java Strings

Program-51

Create a class containing a method to create the mirror image of a String. The method should return the two Strings separated with a pipe(|) symbol .

Method Name	<i>getImage</i>
Argument	<i>String</i>
Return Type	<i>String</i>
For Example	<i>Input : EARTH Output : EARTH HTRAE</i>

Program-52

Create a method to check if a number is an increasing number

Method Name	<i>checkNumber</i>
Argument	<i>String number</i>
Return Type	<i>boolean</i>
For Example	<i>134468 is an increasing number A number is said to be an increasing number if no digit is exceeded by the digit to its left.</i>

Program-53

Create a method that accepts a number and modifies it such that the each of the digit in the newly formed number is equal to the difference between two consecutive digits in the original number. The digit in the units place can be left as it is.

Note: Take the absolute value of the difference. Ex: 6-8 = 2

Method Name	<i>modifyNumber</i>
Algorithm	<i>1. Convert number into String 2. Extract each char using charAt method Convert char to int and find the difference 3. Create new StringBuffer object and keep adding the difference 4. Finally convert StringBuffer to int</i>
Argument	<i>int number1</i>
Return Type	<i>int</i>
For example.	<i>Input: 45862 Output:13242</i>

Program-54

You are asked to create an application for registering the details of jobseeker. The requirement is:

Username should always end with **_job** and there should be atleast minimum of 8 characters to the left of **_job**. Write a function to validate the same. Return true in case the validation is passed. In case of validation failure return false.

Method Name	<i>validateUserName</i>
Argument	<i>String userName</i>
Return Type	<i>boolean</i>

Program-55

Create a method which can perform a particular String operation based on the user's choice. The method should accept the String object and the user's choice and return the output of the operation.

Options are

A: Add the String to itself

B: Replace alternate positions with *

C: Remove duplicate characters in the String

D: Change alternate characters to upper case

Method Name	<i>changeString</i>
Argument	<i>String string, char ch</i>
Return Type	<i>String</i>

Program-56

Create a method that accepts a String and checks if it is a positive string. A string is considered a positive string, if on moving from left to right **each** character in the String comes after the previous characters in the Alphabetical order.

For Example

ANT is a positive String (Since T comes after N and N comes after A)

APPLE is not positive since L comes before P in the alphabetical order.

The method should return true if the entered string is positive

Method Name	<i>checkPositive</i>
Argument	<i>String</i>
Return Type	<i>boolean</i>
Algorithm	<i>Step 1: Convert to Char array. Step 2: Iterate through array, subtract 1st two characters (A-N). This will give the ASCII difference Step 3: If result is negative, then return false and break. Else continue to next loop</i>

Program-57

A company transmits its String data over the network as encrypted data. The encryption logic is as shown given below.

For a String ad the logic is as given

A=a+9=j

D=d+9=m

So the encrypted word would be jm.

Note

If on addition of 9 results in a char greater than z (ASCII value 122) do the encryption in a cyclic manner starting from a. So if any character is 'z' it should be (z+9) which is equal to 127>122. In this case the character would be 9 character starting from 'a' which 'i' so for adz the encrypted value should be adi

Method Name	<i>encryptString</i>
Argument	<i>String</i>
Return Type	<i>String</i>

Program-58

Write a method which accepts a String and moves all the lower case 'a' to the beginning of the String.

Method Name	<i>rearrangeCharacters</i>
Argument	<i>String</i>
Return Type	<i>String</i>
Algorithm	<i>1. Convert the string to a character array 2. Create a StringBuffer object 3. Create a variable(count) to store the number of 'a' present 4. Iterate over the character array and if the character is 'a' increment count for 'a' else add the character to the StringBuffer object. 5. Finally insert the count number of 'a' to the beginning of the StringBuffer object</i>

Program-59

Write a method which accepts an email in a form of String and validate that email values based on the following rules

1) There should be only one @

2) There should be one com

3) There should be only 5 characters between @ and .

Method Name	<i>validateEmail</i>
Argument	<i>String</i>
Return Type	<i>Boolean</i>

Program-60

Write a method which accepts two string parameters, find out the count the number of occurrence of second parameter in the first parameter

Method Name	<i>countWord</i>
Argument	<i>String and String</i>
Return Type	<i>Int</i>
Example	<i>First parameter: hello world, world is wonderful</i>

	Second parameter: world Output:2
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Program-61

Write a method which accepts one string parameter; check whether the first word and the last word in the string match, if they match return the number of characters in the word else return sum of the characters in both

Method Name	checkFirstAndLast
Argument	String
Return Type	int
Example	Input: hello world hello Output:5 Input: hello this world Output:10

Program-62

Write a method which accepts take two input strings and find whether the second input string has all characters within the first input string(it should also include duplicated)

Method Name	sameCharacter
Argument	String and String
Return Type	Boolean
Example	Input: String 1: eat String 2: tea Output: true

Program-63

Write a method which accepts a string and you have to search for x(small x) in the string and place all the x in the back of the string (case sensitive)

Method Name	replaceX
Argument	String
Return Type	String
Example	Input: xAabxbX Output: AabbXxx

Program-64

Write a method which accepts a string and an integer(n), return a string that contains 'n' characters 'n' times

Method Name	countCharacter
Argument	String and int
Return Type	String
Example	input: String is "Hello"

	int value is 2 Output: Lolo
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Program-65

Write a method which accepts a string with the combination of characters and digits, find out sum of digits in string

Method Name	sumOfDigits
Argument	String
Return Type	Int
Example	Input1: ca21 Output: 2+1=3

Program-66

Define a method which accepts 2 values as strings and returns a weaved string.

Write the method with the following specifications:

Name of method *getWeavedString()*

Arguments: *two arguments of type strings*

Return Type: *a string value*

Specifications: *The value returned by the method getWeavedString() is determined by the following rules:*

Values must not be blank. If yes, then return -1 as string.

If value1 is greater than value2 in length, then return a concatenated string which looks like value2+value1+value2.

If value1 is smaller than value2 in length, then return a concatenated string which looks like value1+value2+value1.

If value1 equal to value2 in length, then return a concatenated string which contains each character from both the values at the same position.

For example,

"Hello" "Hello" result = "HHeelloo"

Program-67

Define a method which returns "FIZZ" if the input is a multiple of 3 prints "BIZZ" if the input is a multiple of 5 and prints "FIZZBIZZ" if the input is a multiple of 3 and 5. Print the given input if it is multiple of neither 3 nor 5.

Write the method with the following specifications:

Name of method : *getFizzBizz()* // which accepts an integer value as argument and return the sum of it's digits.

Arguments: *one argument of type integer*

Return Type: *string*

Specifications: *The value returned by the method getFizzBizz() is determined by the following rules:*

If the input is a multiple of 3 return "FIZZ" (Ex:- Input 33 output FIZZ)
If the input is a multiple of 5 return "BIZZ" (Ex:- Input 5 output BIZZ)
If the input is a multiple of 3 and 5 return "FIZZBIZZ" (Ex:- Input 15 output FIZZBIZZ)
if the input is not a multiple of 3 or 5, return the number itself in the form of string
If the given value is negative or zero, return "Error"

Program-68

Write code for encrypting given text

Method details as follows

Name of Method: `encrypt()` // accepts a string and return encoded string.

Arguments: one String

Return type: one String

Method should meet the following functional requirements

1. Return encoded string for a given string
2. Return null, if input contains anything other than a-z.

Explanation:

Plain Text string is "abcdefghijklmnopqrstuvwxyz";

Cipher Text string is "zyxwvutsrqponmlkjihgfedcba";

if the input is "abc", that need to be changed to "zyx".

index of "a" in plain text is equal to index of "z" in cipher text

index of "b" is plain text equal to index of "y" in cipher text and so on.

Plain text character from the given string should be changed to cipher text character with the same index.

Program-69

Write code for decrypting given text

Method details as follows

Method name : `decrypt()` // accepts a string and returns plain text

Arguments : one String

Return type : one String

Method should meet the following functional requirements

1. Return decoded string for a given string
2. Return null, if input contains special characters or digits

Explanation:

Plain Text string is "abcdefghijklmnopqrstuvwxyz";

Cipher Text string is "zyxwvutsrqponmlkjihgfedcba";

if the input is "zyx", that need to be changed to "abc".

Input: zyx

Output: abc

index of "z" in cipher text is equal to index of "a" in plain text

index of "y" is cipher text equal to index of "b" in plain text and so on.

Cipher text character from the given string should be changed to plain text character with the same index.

Program-70

Write logic for reversing each word in a given string.

Define following method

Name of Method : *reverse*

Arguments : *String*

Return type : *String*

Input : *talent*

Output : *tnelat*

Input : *talent sprint*

Output : *anelat tnirps*

Method should meet the functional expectations.

1. Return string that contains each word in reverse order.

2. Return null, if input is null.

3. Return null, if input string is empty.

Program-71

Write logic for moving character of a string from left to right and last character to first position.

Characters of string need to be moved by given no of positions.

Define following method

Name of Method : *rotate*

Arguments : *String, integer (i. e no_of_positions)*

Return type : *String*

Input : *talent 1*

Output : *ttalen*

Input : *talent 2*

Output : *nttale*

Input : *talent 3*

Output : *enttal*

Method should meet the functional expectations

1. Move characters of string by given no of positions and return changed string

2. Return null, if input is null

3. Return same string, if no_of_positions is zero or Negative.

4. if no.of positions is exceeding no.of characters in string then return same string.